

HW# 32

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CS-F

QNo 1

$$A^t A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A^t b = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 6 \\ 5 \\ 3 \end{bmatrix}$$

$$\hat{x} = ?$$

$$\left[\begin{array}{ccc|c} 3 & 2 & 1 & 6 \\ 2 & 2 & 1 & 5 \\ 1 & 1 & 1 & 3 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 2 & 1 & 5 \\ 3 & 2 & 1 & 6 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & -1 & -1 \\ 0 & -1 & -2 & -3 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & +1 & 0 & 2 \\ 0 & 0 & +1 & 1 \end{array} \right] \Rightarrow \begin{aligned} \hat{x}_1 &= 1 \\ \hat{x}_2 &= 1 \\ \hat{x}_3 &= 1 \end{aligned}$$

$$\hat{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$e = b - A\hat{x}$$

$$e = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$e = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\|e\| = 0$$



Qn2

$$Q = \begin{bmatrix} 1/\sqrt{3} & -\sqrt{6}/3 & 0 \\ 1/\sqrt{3} & 1/\sqrt{6} & -1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{6} & 1/\sqrt{2} \end{bmatrix}$$

$$R = \begin{bmatrix} \sqrt{3} & 2/\sqrt{3} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{6} \\ 0 & 0 & 1/\sqrt{2} \end{bmatrix}$$

$$R\hat{x} = Q^T b$$

$$Q^T b = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{6} & 1/\sqrt{2} \\ -\sqrt{6}/3 & 1/\sqrt{6} & 1/\sqrt{2} \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2\sqrt{3} \\ \sqrt{6}/2 \\ \sqrt{2}/2 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} \sqrt{3} & 2/\sqrt{3} & 1/\sqrt{3} & 2\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{6} & \sqrt{6}/2 \\ 0 & 0 & 1/\sqrt{2} & \sqrt{2}/2 \end{array} \right] \sim \left[\begin{array}{ccc|c} \sqrt{3} & 2/\sqrt{3} & 1/\sqrt{3} & 2\sqrt{3} \\ 0 & 2 & 1 & 3 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

$$\begin{aligned} \hat{x}_3 &= 1 \\ \hat{x}_2 &= 1 \\ \hat{x}_1 &= 1 \end{aligned} \quad \hat{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

QNo 3

$$A^+ = (A^T A)^{-1} A^T$$

$$= \begin{bmatrix} 3 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$A^+ A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Q No 4

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$A^+ = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

For invertible matrices, pseudo inverse is equal to inverse

Qno 5

$$\begin{aligned}
 P &= A (A^T A)^{-1} A^T \\
 &= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

Qno 6

~~AQR~~

~~$$\begin{aligned}
 P &= QR (QR)^T (QR)^{-1} (QR)^T \\
 &= QR (R^T Q^T Q R)^{-1} R^T Q^T \\
 &= QR (I) R^T Q^T \\
 &= Q I Q^T \\
 &= QQ^T
 \end{aligned}$$~~

$$\hat{x} = (A^T A)^{-1} A^T b$$

$$= \begin{bmatrix} 5/84 & -1/84 \\ -1/84 & 17/84 \end{bmatrix} \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5/84 & -1/84 \\ -1/84 & 17/84 \end{bmatrix} \begin{bmatrix} 9 \\ 1 \end{bmatrix}$$

$$= \frac{1}{84} \begin{bmatrix} 5 & -1 \\ -1 & 17 \end{bmatrix} \begin{bmatrix} 9 \\ 1 \end{bmatrix}$$

$$= \frac{1}{84} \begin{bmatrix} 44 \\ 8 \end{bmatrix}$$

$$\hat{x} = \begin{bmatrix} 11/21 \\ 2/21 \end{bmatrix}$$

$$\hat{b} = A \hat{x}$$

$$= \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 11/21 \\ 2/21 \end{bmatrix}$$

$$\hat{b} = \begin{bmatrix} 44/21 \\ 4/21 \\ 13/21 \end{bmatrix}$$

5.4)

$$v_1 = \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} - \frac{1}{17} \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} -4/17 \\ 2 \\ 16/17 \end{bmatrix}$$

$$v_1 \Rightarrow u_1 = \begin{bmatrix} 4/\sqrt{17} \\ 0 \\ 1/\sqrt{17} \end{bmatrix}$$

$$u_2 = \begin{bmatrix} -4 \\ 34 \\ 16 \end{bmatrix} \frac{1}{\sqrt{1428}}$$

$$\begin{bmatrix} 4/\sqrt{17} & 0 & 1/\sqrt{17} \\ -4/\sqrt{1428} & 34/\sqrt{1428} & 16/\sqrt{1428} \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} = R$$

$$R = \begin{bmatrix} \sqrt{17} & 1/\sqrt{17} \\ 0 & 34/\sqrt{1428} \end{bmatrix}$$

$$Q = \begin{bmatrix} 4/\sqrt{17} & -4/\sqrt{1428} \\ 0 & 34/\sqrt{1428} \\ 1/\sqrt{17} & 16/\sqrt{1428} \end{bmatrix}$$